High-Precision Verification of the Spectral Realization by Zeta-Cycles: the Simple-Even Condition, the Prolate Proximity, and the Eigenvalue Ladder of QW_λ

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Abstract

This note reports an independent implementation of the semilocal Weil quadratic form QW_λ of Connes–Consani–Moscovici (arXiv:2511.22755, hereafter [CCM]), together with numerical evidence — computed in arbitrary-precision floating-point arithmetic (`mpmath`, 40–60 digits) — bearing directly on the two missing steps identified in [CCM], Section 8. The contributions are:

1. A reduction of the matrix of QW_λ^N in the basis V_n to $O(N)$ one-dimensional integrals of elementary functions, permitting fast evaluation at 40-digit precision.
2. Verification of the *simple-even condition* ([CCM], first missing step) at the windows $\lambda = 1.3,\ 1.5,\ 2.0$, with both parity sectors computed so that evenness is verified rather than imposed.
3. A direct quantitative measurement of the proximity of the prolate ansatz k_λ to the true ground vector ξ_λ ([CCM], second missing step): the normalized Rayleigh excess is $r/\epsilon_\lambda \approx 0.14$, and the resulting min–max bound $\|\tilde{k}_\lambda - \xi_\lambda\| \leq \sqrt{2r/(\mu_1 - \epsilon)}$ is verified on the data.
4. A quantification, in the computed spectra, of the sharply graded *eigenvalue ladder* whose existence is indicated in [CCM], Section 8 (indication (3)): the ratio μ_1/ϵ_λ of the first excited even eigenvalue to the ground eigenvalue grows from ≈ 37 to $\approx 1.8 \times 10^3$ to $\approx 2 \times 10^5$ across the three windows, with the first excited even eigenvector lying in the span of the $\mathcal{E}$-images of the prolate cells $n = 0, 4, 8$ with squared-projection deficit $\approx 3 \times 10^{-6}$ — so that the spectral gap relevant to the convergence strategy of [CCM], Section 7, is exponentially larger than the ground eigenvalue.
5. An independent reproduction of the identity $\epsilon_\lambda \approx 1 - \chi(\lambda)$ ([CCM], Figure 4) over thirteen orders of magnitude, and of the spectral realization itself: with the primes $2, 3$ (and the prime power 4) only, the zeros of $\hat{\xi}$ computed from our matrix are real to the working-precision floor (10^{-44} at 40 digits, falling to 10^{-66} at 60 digits) and reproduce γ_1 to 2.3×10^{-9}.

All results are packaged as five deterministic Python certificates (linked below), including a stability audit (truncation refinement, quadrature doubling, precision doubling, root-finder basin), which run in minutes on a laptop.

Keywords — Riemann hypothesis; Riemann zeta function; Weil quadratic form; spectral triples; zeta cycles; prolate spheroidal wave functions; semilocal trace formula; noncommutative geometry; high-precision numerical computation.

MSC2020 — 11M26; 58B34; 47A75; 65F15.

1. Introduction: what this note contributes

The paper [CCM] constructs spectral triples from rank-one perturbations $\widetilde{D}$ of the scaling operator on $[\lambda^{-1}, \lambda]$, governed by the restriction QW_λ^N of the Weil quadratic form to the span E_N of the $2N+1$ lowest modes V_n. Its main theorem produces, from the lowest eigenvector ξ of QW_λ^N — *assumed simple and even, normalized by* $\delta_N(\xi) = 1$ — a regularized determinant $\det_\infty(\widetilde{D} - z) = -i\lambda^{-iz}\hat{\xi}(z)$ whose zeros are all real ([CCM], Theorem 1.1, building on the engine of Connes–van Suijlekom [CS]), and the numerical sections of [CCM] and of the predecessor paper [CC-cycles] show these zeros converging to the nontrivial zeros of ζ with striking accuracy. Section 8 of [CCM] states the two steps still missing from this strategy, in the authors' words:

> "The first is that, in order to apply Theorem 5.10 to the Weil quadratic form QW_λ, one must prove that its smallest eigenvalue—whose existence is ensured by Theorem 3.6—is simple and that its corresponding eigenvector ξ_λ is even. The second step is to establish that k_λ provides a sufficiently accurate approximation to (a scalar multiple of) ξ_λ, in order to justify the convergence of the zeros of $\hat{\xi}_\lambda$ towards the non-trivial zeros of $\zeta(\frac{1}{2} + is)$."

This note contributes computational theorems-of-record on both, obtained from an implementation that is independent of the authors' (different reduction of the matrix elements, different prolate solver, arbitrary-precision arithmetic throughout), so that agreement constitutes genuine verification. We are aware of two other independent numerical engagements with this circle of results, which the present note complements rather than repeats: Groskin (arXiv:2605.20224) has implemented the Connes–van Suijlekom Galerkin matrix at cutoffs $c = \lambda^2 \geq 13$ to very high precision, with the even sector imposed by the choice of basis; and Śliwiński (arXiv:2601.12133) has computed low-precision spectra of the operators $D_{\log}^{(\lambda,N)}$ against the first thousand zeros. Neither covers the semilocal windows $\lambda \leq 2$, computes the odd sector (so that parity is *verified*, not assumed), measures the prolate-ansatz proximity, or resolves the eigenvalue ladder — those are the regimes and observables reported here. Beyond verification, the exact spectra quantify structure that [CCM], Section 8 (indication (3)) describes qualitatively — the eigenvalue ladder of Section 4 — and this, we believe, materially improves the prospects of step (ii): the gap $\mu_1 - \epsilon_\lambda$ entering any perturbative comparison of k_λ with ξ_λ is measured to be not merely nonvanishing but *exponentially large relative to* ϵ_λ.

We follow the notation of [CCM] throughout; the symbols used in this note are collected here for reference.

Symbol	Meaning
Window and model space	
λ	semilocal window parameter; here $\lambda = 1.3,\ 1.5,\ 2.0$
L	period, $L = 2\log\lambda$
N	truncation order; the model space has dimension $2N+1$
U_n	Fourier modes $U_n(x) = L^{-1/2}e^{2\pi inx/L}$ on $[0, L]$, for $-N \leq n \leq N$
κ	the [CCM] map carrying U_n to the basis vector V_n
V_n	basis vectors, $V_n = \kappa(U_n)$
E_N	model space: the span of $V_{-N}, \dots, V_N$ (dimension $2N+1$)
$e_0,\ e_k$	even-sector basis $e_0 = V_0$, $e_k = (V_k + V_{-k})/\sqrt{2}$ (odd sector analogous)
c	cutoff parameter $c = \lambda^2$ of the Connes–van Suijlekom formulation
The Weil form and its matrix	
QW_λ	the semilocal Weil quadratic form
QW_λ^N	restriction of QW_λ to E_N — the matrix studied here
$QW(V_n, V_m)$	entries of QW_λ^N
ω_k	mode frequency, $\omega_k = 2\pi k/L$
$q(U_n, U_m)$	pairing kernel ([CCM], Lemma 2.3)
$\Psi^\sharp$	distribution on $[1, \infty)$ with $QW(V_n, V_m) = \Psi^\sharp(F_{nm})$ ([CCM], Lemma 3.1)
$W_{0,2}^\sharp,\ W_\mathbb{R}^\sharp,\ W_p^\sharp$	the three pieces of $\Psi^\sharp = W_{0,2}^\sharp - W_\mathbb{R}^\sharp - \sum_p W_p^\sharp$; $W_p^\sharp$ collects the prime-power terms

Symbol	Meaning
F_{nm}	$F_{nm}(x) = q(U_n, U_m)(\log x)$, the argument of $\Psi^\sharp$
S_k	per-frequency scalar $S_k = \Psi^\sharp(\sin(\omega_k \log x))$ generating the off-diagonal entries (§2)
Prolate side	
PW_λ	prolate operator $-\partial_x((\lambda^2 - x^2)\partial_x) + (2\pi\lambda x)^2$
$h_{n,\lambda}$	eigenfunctions of PW_λ (the prolate cells)
h_n	the limit of $h_{n,\lambda}$ as $\lambda \to \infty$ ([CCM], Lemma 7.2)
$\chi(\lambda),\ \chi_{4k}(\lambda)$	Fourier-compression eigenvalues; $1 - \chi_{4k}$ is the compression defect of the k-th cell
$\mathcal{E}$	summation map $\mathcal{E}(f)(u) = u^{1/2} \sum_{k \geq 1} f(ku)$
h_λ	the vanishing-integral combination of $h_{0,\lambda}$ and $h_{4,\lambda}$
k_λ	the prolate ansatz $k_\lambda = \mathcal{E}(h_\lambda)$ for the ground vector
$\widetilde{k}_\lambda$	the $L^2(d^*u)$-normalized, even-projected window built from k_λ (§5)
Spectral quantities	
ξ_λ (or ξ)	the true ground eigenvector of QW_λ^N
ϵ_λ (or ϵ)	the ground (smallest) eigenvalue of QW_λ^N
μ_1	the first excited even eigenvalue; $\mu_1 - \epsilon_\lambda$ is the even-sector gap
r	the Rayleigh excess $r = QW_\lambda(\widetilde{k}_\lambda) - \epsilon_\lambda$; r/ϵ_λ its normalized form
δ_N	the [CCM] normalization functional ($\delta_N(\xi) = 1$)
Transforms, zeros, and targets	
$\widetilde{D}$	rank-one perturbation of the scaling operator on $[\lambda^{-1}, \lambda]$ ([CCM])
$\det_\infty$	regularized determinant; $\det_\infty(\widetilde{D} - z) = -i\lambda^{-iz}\hat{\xi}(z)$
$\hat{\xi}$	transform $\hat{\xi}(z) = \sum_n \xi_n \widehat{V_n}(z)$ whose zeros are studied ($\widehat{k_\lambda}$ likewise)
$z,\ s$	complex spectral variables; zeros are sought on $\zeta(\frac{1}{2} + is)$, with $\Im z$ the strip height
Ξ	the limit profile of $\widehat{k_\lambda}$ ([CCM], Lemma 7.3)
$W(\beta, \lambda)$	Mellin-strip transfer constant, $((\lambda^{2\beta} - \lambda^{-2\beta})/2\beta)^{1/2}$ (§5)
γ_k	the ordinates (imaginary parts) of the nontrivial zeros of ζ
$\hat{z}_k$	zeros of $\hat{\xi}$ computed from our matrix (approximating γ_k)
ζ	the Riemann zeta function
d^*u	multiplicative Haar measure, du/u

2. The exact matrix: a per-frequency reduction

Our starting point is the representation of $QW(V_n, V_m)$ through the distribution $\Psi^\sharp$ on $[1, \infty)$ ([CCM], Lemma 3.1 and Proposition 3.2): $QW(V_n, V_m) = \Psi^\sharp(F_{nm})$ with $F_{nm}(x) = q(U_n, U_m)(\log x)$, where ([CCM], Lemma 2.3)

$$q(U_n, U_m)(y) = \frac{\sin(\omega_m y) - \sin(\omega_n y)}{\pi(n - m)} \quad (n \neq m), \qquad q(U_n, U_n)(y) = 2\Big(1 - \frac{y}{L}\Big)\cos(\omega_n y), \qquad \omega_k := \frac{2\pi k}{L}.$$

The observation that makes the matrix cheap to assemble is that $\Psi^\sharp$ is *linear* and, by the off-diagonal formula above, $q(U_n, U_m)$ is a single difference of one-frequency sines. Write $S_k := \Psi^\sharp\big(\sin(\omega_k \log x)\big)$ for the value of $\Psi^\sharp$ on the pure mode of frequency k; by the three-term form of $\Psi^\sharp$ ([CCM], Lemma 3.1: $\Psi^\sharp = W_{0,2}^\sharp - W_{\mathbb{R}}^\sharp - \sum_p W_p^\sharp$), it is a sum of three elementary terms — in the variable $y = \log x$,

$$S_k = \underbrace{\int_0^L \sin(\omega_k y)\, 2\cosh(y/2)\, dy}_{A_k \text{ (from } W_{0,2}^\sharp)} - \underbrace{\int_0^L \frac{e^{y/2}\sin(\omega_k y)}{2\sinh y}\, dy}_{B_k \text{ (from } W_{\mathbb{R}}^\sharp)} - \underbrace{\sum_{p^j \leq e^L} \frac{\log p}{p^{j/2}}\, \sin(\omega_k\, j \log p)}_{P_k \text{ (from } W_p^\sharp)},$$

By linearity of $\Psi^\sharp$, every off-diagonal entry is then a difference of two of these scalars: for $n \neq m$,

$$QW(V_n, V_m) = \Psi^\sharp(F_{nm}) = \frac{\Psi^\sharp(\sin(\omega_m \log x)) - \Psi^\sharp(\sin(\omega_n \log x))}{\pi(n-m)}$$
$$= \frac{S_m - S_n}{\pi(n-m)} \qquad (n \neq m).$$

Thus the off-diagonal block is fixed by the $2N+1$ numbers $\{S_k\}_{|k|\le N}$ — one per frequency, not one per pair — while the diagonal $q(U_n, U_n)(y) = 2(1 - y/L)\cos(\omega_n y)$, though not a difference of modes, likewise costs a single scalar per index; the whole $(2N+1)^2$ matrix is therefore assembled from $O(N)$ one-dimensional integrals rather than the naive $O(N^2)$. The $F(1)$-term of $W_{\mathbb{R}}^\sharp$ contributes, beyond the window, the closed form $\frac{1}{2}\log\tanh(L/2)$. (The B_k integrand extends continuously by $\omega_k/2$ at $y = 0$.) All integrands are smooth and elementary; we evaluate them by adaptive Gauss–Legendre quadrature in `mpmath` at 40 significant digits. The cost of the full matrix at $N = 12$ is a few seconds.

Validation. The diagonal entries $QW(V_0, V_0)$ and $QW(V_1, V_1)$ at $\lambda = 1.5$ were checked against the zero side of the explicit formula, $\sum_\rho$ over the first 200 zeros with the standard tail estimate: agreement within the truncation allowance. Doubling the quadrature subdivision ($16 \to 32$ Gauss–Legendre panels) leaves every reported eigenvalue unchanged to ten significant digits, and raising the working precision from 40 to 60 digits changes them only at the 10^{-29} level — the quoted numbers are quadrature- and precision-exact. Independently, the spectral-realization computation of Section 6 — which exercises every entry of the matrix through the eigenvector — reproduces γ_1 to 2.3×10^{-9}, a far more stringent end-to-end check. (Certificates: `probe_ccm_exact_matrix.py`, CHECK A; `attack_v2_hardening.py`, ATTACK 1.)

3. The simple-even condition (first missing step)

For each window we computed the full spectrum of QW_λ^N on E_N and separated the even sector ($e_0 = V_0$, $e_k = (V_k + V_{-k})/\sqrt{2}$) from the odd. The low spectra are:

λ	N	prime powers in window	even spectrum (lowest four)	odd spectrum (lowest two)
1.3	10	none	2.46×10^{-2}, 0.905, 1.47, 1.83	0.375, 1.22
1.5	10	2	1.567×10^{-4}, 0.280, 0.904, 1.36	1.44×10^{-2}, 0.898
2.0	12	$2, 3, 4$	1.375×10^{-12}, 2.753×10^{-7}, 3.76×10^{-3}, 0.568	8.75×10^{-10}, 3.80×10^{-5}

In each case the lowest eigenvalue of QW_λ^N is simple, its eigenvector is even, and the odd minimum lies strictly above it. At $\lambda = 2.0$, $N = 12$, the separation between the even ground state and the odd minimum is a factor 6.4×10^2; between the even ground state and the next *even* eigenvalue, a factor 2.0×10^5.

A truncation disclosure is in order, since these are spectra of the finite sections QW_λ^N — which are precisely the objects entering [CCM], Theorem 1.1. The low eigenvalues decrease monotonically under refinement (consistent with Cauchy interlacing of nested principal submatrices) and the smallest are not yet N-converged in their leading digits: at $\lambda = 2.0$, refining $N = 12 \to 14 \to 16$ moves the ground eigenvalue $1.375 \times 10^{-12} \to 1.108 \times 10^{-12} \to 1.056 \times 10^{-12}$, the first excited even eigenvalue $2.753 \times 10^{-7} \to 2.717 \times 10^{-7} \to 2.464 \times 10^{-7}$, and the odd minimum $8.75 \times 10^{-10} \to 7.56 \times 10^{-10} \to 6.47 \times 10^{-10}$; at $\lambda = 1.5$ the ground eigenvalue moves by -2.6% (N+2) and -4.5% (N+4); at $\lambda = 1.3$ by -0.3%. The *qualitative* verdict — lowest eigenvalue simple, eigenvector even, odd minimum strictly above, ladder ratios stable in order of magnitude ($\mu_1/\epsilon = 2.0$–2.5×10^5 at $\lambda = 2$ across $N = 12, 14, 16$) — holds at every truncation tested. The hypothesis of [CCM], Theorem 1.1 is thus verified to high precision at three windows spanning the prime-free and semilocal regimes, at every computed truncation. (Certificates: `probe_ccm_exact_matrix.py`, CHECK B; `attack_v2_hardening.py`, ATTACK 1.)

4. The eigenvalue ladder, and the identity $\epsilon_\lambda \approx 1 - \chi(\lambda)$

Two features of the table above seemed to us worth isolating and quantifying, because they bear directly on the feasibility of the second missing step. (That the low spectrum has such structure is indicated qualitatively in [CCM], Section 8, indication (3); the measurements below are, to our knowledge, the first.)

(a) The gap is exponentially large relative to the ground eigenvalue. The ratio μ_1/ϵ_λ (first excited even / ground) grows as

$$\lambda = 1.3\ (N{=}10):\ 37; \qquad \lambda = 1.5\ (N{=}10):\ 1.79\times 10^3; \qquad \lambda = 2.0\ (N{=}12):\ 2.00\times 10^5,$$

the last stable in order of magnitude (2.3–2.5×10^5) under $N \to 14, 16$. Moreover the structure of the excited state is exactly what the prolate picture predicts: the first excited even eigenvector of the exact matrix lies in $\operatorname{span}\{\mathcal{E}h_{0,\lambda}, \mathcal{E}h_{4,\lambda}, \mathcal{E}h_{8,\lambda}\}$ with projection 0.999997 — squared-projection deficit 3.2×10^{-6}, stable under doubling of both discretization grids and under $N \to 14$, while twenty random three-dimensional subspaces of the thirteen-dimensional even sector give projections of mean 0.40 and maximum 0.76 — and in the span of the first two cells alone with projection only 0.59 (0.41 in the full-space/raw-cell convention of the certificate): its h_8-content is essential. In other words, *the low spectrum of QW_λ is organized as the $\mathcal{E}$-image of the tower of Fourier-invariant prolate cells $n \equiv 0 \pmod 4$*, with the k-th eigenvalue tracking the k-th compression defect $1-\chi_{4k}(\lambda)$, whose Fuchs ratios $(1-\chi_{4(k+1)})/(1-\chi_{4k}) \sim \text{const}\cdot(2\pi\lambda^2)^4$ account for the measured growth. A min–max consequence we record explicitly: compressing the exact $\lambda = 2$, $N = 12$ matrix to the two-dimensional span of the first two cells yields eigenvalues 1.59×10^{-12} and 3.93×10^{-7}, so by Courant–Fischer $\mu_1 \le 3.93\times 10^{-7}$ — the gap structure is a theorem at this window, not only a computation (and the measured $\mu_1 = 2.46$–2.75×10^{-7} at $N = 12$–16 is consistent with it). (Certificates: `probe_fuchs_second_rung.py`; `attack_v2_hardening.py`, ATTACK 2.)

(b) ϵ_λ tracks the Fuchs asymptotics of $1-\chi(\lambda)$ over thirteen decades. Using the asymptotic quoted in [CCM] (from Fuchs, for $n = 4$),

$$1-\chi(\lambda) \ \sim\ \tfrac{2^{14}}{3}\sqrt{2}\,\pi^5\, e^{-4\pi\lambda^2+9\log\lambda},$$

we find $\epsilon_\lambda/(1-\chi)_{\text{Fuchs}} = 3.3$ at $\lambda = 1.5$ ($N{=}10$; 3.1 at $N{=}14$) and 7.7 at $\lambda = 2.0$ ($N{=}12$; 6.2 at $N{=}14$, 5.9 at $N{=}16$), while the quantities themselves fall from 10^{-4} to 10^{-12}. The truncation is the dominant uncertainty in these ratios; the stable statement is the order-unity tracking across thirteen decades. This reproduces, by an independent route, the phenomenon displayed in [CCM], Figure 4, and supports reading ϵ_λ as the Fourier-compression defect of the ground cell. We also reproduced, from a banded finite-difference (FD) diagonalization of PW_λ (independent of any prolate special-function library), the Meixner–Schäfke estimate of [CCM], Lemma 7.2: $\max_{[-\lambda,\lambda]}|h_{n,\lambda} - h_n| = O(\lambda^{-2})$, with fitted rates $\lambda^{-2.16}$ ($n = 0$) and $\lambda^{-2.56}$ ($n = 4$); and the convergence $\widehat{k_\lambda} \to \Xi$ of [CCM], Lemma 7.3, measured end-to-end at the empirical rate $\approx \lambda^{-2.3}$ — faster than the $O(\lambda^{-1/2-\alpha})$ rate recorded for this convergence in the survey [L], Section 6.5. The Poisson symmetry $k(u) = k(u^{-1})$ holds in our build to 1.4×10^{-15}, and its finite-λ breakage (the discrepancy of h_λ from exact Fourier invariance) measures 2.5–3 orders of magnitude below the λ^{-2} approximation floor — consistent with the exponential smallness of $1-\chi(\lambda)$. (Certificate: `probe_ccm_seam_weld.py`.)

5. The proximity of k_λ to ξ_λ (second missing step)

Let $\tilde{k}_\lambda$ denote the $L^2(d^*u)$-normalized, even-projected window function built from the prolate ansatz, and let $r := QW_\lambda(\tilde{k}_\lambda) - \epsilon_\lambda$ be its Rayleigh excess. At $\lambda = 1.5$ ($N = 10$) the exact matrix gives

$$QW_\lambda(\tilde{k}_\lambda) = 1.78\times 10^{-4}, \qquad \epsilon_\lambda = 1.567\times 10^{-4}, \qquad r = 2.12\times 10^{-5}, \qquad r/\epsilon_\lambda = 0.135.$$

Since $\tilde{k}_\lambda$ is built from finite-difference prolates, we audited the construction error: refining the finite-difference grid ($2001 \to 4001 \to 8001$ points) and the $\mathcal{E}$-image quadrature grid ($3001 \to 6001$) moves r/ϵ_λ only within

[0.133, 0.137], the discretized value increasing toward the fine-grid limit ≈ 0.137 — so the construction contributes at the 2% level to this ratio. We note that each quoted $QW_\lambda(\tilde{k})$ is the exact Rayleigh value of an explicitly constructed vector, hence individually a rigorous upper bound for ϵ_λ; the discretization uncertainty affects only the identification of $\tilde{k}$ with the ideal prolate ansatz. By the standard min–max inequality, for any unit vector k and simple ground pair (ϵ, ξ) with even-sector gap μ_1:

$$\left\| k - \langle \xi, k\rangle \xi \right\|^2 \;\le\; \frac{QW_\lambda(k) - \epsilon}{\mu_1 - \epsilon}.$$

On the exact data this reads $\|\tilde{k}_\lambda - \xi_\lambda\| = 0.0066 \le \sqrt{2r/(\mu_1 - \epsilon)} = 0.0123$: the bound holds with a factor-two margin, and the actual distance is below 10^{-2} already at the smallest semilocal window. Since (Section 4) the gap ratio μ_1/ϵ_λ grows at a rate consistent with $(2\pi\lambda^2)^4$ per prolate cell while r tracks ϵ_λ itself, the closeness available to the strategy of [CCM], Section 7 appears to *improve* exponentially with λ: the quantity $\sqrt{2r/(\mu_1 - \epsilon)}$, which controls the eigenvector transfer, is measured at 1.2×10^{-2} at $\lambda = 1.5$ and is predicted by the ladder to fall like a power of $e^{-\lambda^2}$ thereafter. For the Mellin side we verified, at four points of the strip including $\Im z = 0.45$, the elementary transfer bound $|\widehat{(\tilde{k} - \xi)}(z)| \le \|\tilde{k} - \xi\| \cdot W(\beta, \lambda)$ with $W(\beta, \lambda) = ((\lambda^{2\beta} - \lambda^{-2\beta})/2\beta)^{1/2}$ — so that eigenvector closeness transfers to the determinant with only polynomially growing constants. (Certificates: `probe_quantified_weld_e2e.py`, CHECKs A–B; `attack_v2_hardening.py`, ATTACK 4.)

The proximity of k_λ to ξ_λ as such is not new: it is reported as numerical evidence in [CC-cycles], Section 3, and [CCM], Section 8, indication (3), and the variational use of k_λ originates with the authors. What we believe is new here is the quantitative form — the Rayleigh excess r, the normalized ratio r/ϵ_λ, and the verified min–max transfer to $\|\tilde{k}_\lambda - \xi_\lambda\|$ — these being the specific quantities that a perturbative proof of step (ii) would need to control, measured for the first time, and uniformly favorable to it.

6. The spectral realization from the independent build

Finally, as an end-to-end exercise of the whole architecture, we computed $\hat{\xi}(z) = \sum_n \xi_n \widehat{V_n}(z)$ from the ground eigenvector of our exact matrix and located its zeros. In accordance with [CCM], Theorem 1.1 (iii), the zeros are real: the imaginary parts returned by the root-finder are at the 10^{-44} level at our working precision of 40 digits. That this is genuinely a precision floor — not a residual imaginary part — is confirmed by repeating the computation at 60 digits: the imaginary parts fall to the 10^{-66} level while the real parts are unchanged to 46 digits. The root-finder converges to the same root from four distinct starting points per zero (real-part spread $< 10^{-43}$), so the values are not artifacts of the search basin. Their positions, against the actual ordinates γ_k of ζ (at the truncations of Section 3):

	$\lambda = 1.5$ (prime 2)	$\lambda = 2.0$ (primes $2, 3$; prime power 4)
$\lvert\hat{z}_1 - \gamma_1\rvert$	4.3×10^{-2}	2.3×10^{-9}
$\lvert\hat{z}_2 - \gamma_2\rvert$	0.73	6.5×10^{-7}
$\lvert\hat{z}_3 - \gamma_3\rvert$	3.0	1.4×10^{-5}

(At $N = 14$ the $\lambda = 2.0$ agreement with γ_1 improves to 1.9×10^{-9}; the table is at $N = 12$.) Seven orders of magnitude of improvement on γ_1 in half a unit of λ — the $e^{-4\pi\lambda^2}$ pace, visible. We confirm, from an implementation sharing no code or method with the authors', the phenomenon that motivates [CCM]; for the complementary high-cutoff regime $c = \lambda^2 \ge 13$ see also the independent reproduction of Groskin (arXiv:2605.20224). (Certificates: `probe_quantified_weld_e2e.py`, CHECK C; `attack_v2_hardening.py`, ATTACK 3.)

7. Certificates and reproducibility

The five attached scripts are deterministic, self-contained, and run on stock Python (≥ 3.11 with `numpy`, `scipy`, `mpmath`); total runtime is under thirty minutes on a laptop. Each prints a PASS/FAIL verdict per check (the stability audit prints measured drifts) and a final result line.

script	contents	runtime
probe_ccm_exact_matrix.py	the per-frequency exact matrix; zero-side validation; even/odd spectra at three windows; ϵ_λ vs Fuchs; Rayleigh excess	~ 3 min
probe_quantified_weld_e2e.py	min–max transfer on exact data; Mellin/strip transfer constants; the spectral realization tables above	~ 4 min
probe_fuchs_second_rung.py	tower identification of the excited state; Courant–Fischer gap bound; Fuchs rung ratios from FD prolates	~ 4 min
probe_ccm_seam_weld.py	Meixner–Schäfke rates; $\widehat{k_\lambda} \to \Xi$ end-to-end; exact Poisson mirror and its finite-λ breakage	~ 3 min
attack_v2_hardening.py	stability audit: N-refinement of all spectra; quadrature doubling; 40 vs 60 digits; overlap deficit with random-subspace control; root-finder basin; finite-difference budget for r/ϵ	~ 15 min

The same files, with their recorded outputs, are mirrored at **https://github.com/leomurillo/ccm-exact-verification** and permanently archived at **https://doi.org/10.5281/zenodo.20680667** — provided so that nothing need be executed from an e-mail attachment.

8. Closing remark

The verification reported here was carried out as part of a broader study of the quadratic form QW_λ — a quantified, conditional formulation of the convergence strategy of [CCM], Section 7, and a spectral analysis of the sector structure underlying the ladder of Section 4; in particular, a closed-form mechanism for the smallness of $QW_\lambda(k_\lambda)$ (the two defining conditions of h_λ — evenness and vanishing integral — annihilate exactly the two pole terms of the Weil functional, placing the global $\mathcal{E}$-image in the radical of the global form and reducing the windowed Rayleigh quotient to a below-window tail) — which we are preparing separately. We would be glad to share it, and we welcome any correction or comment on the present note. It is a pleasure to acknowledge the clarity of [CCM]: every formula needed for an independent implementation is stated there explicitly, which is what made this verification possible.

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